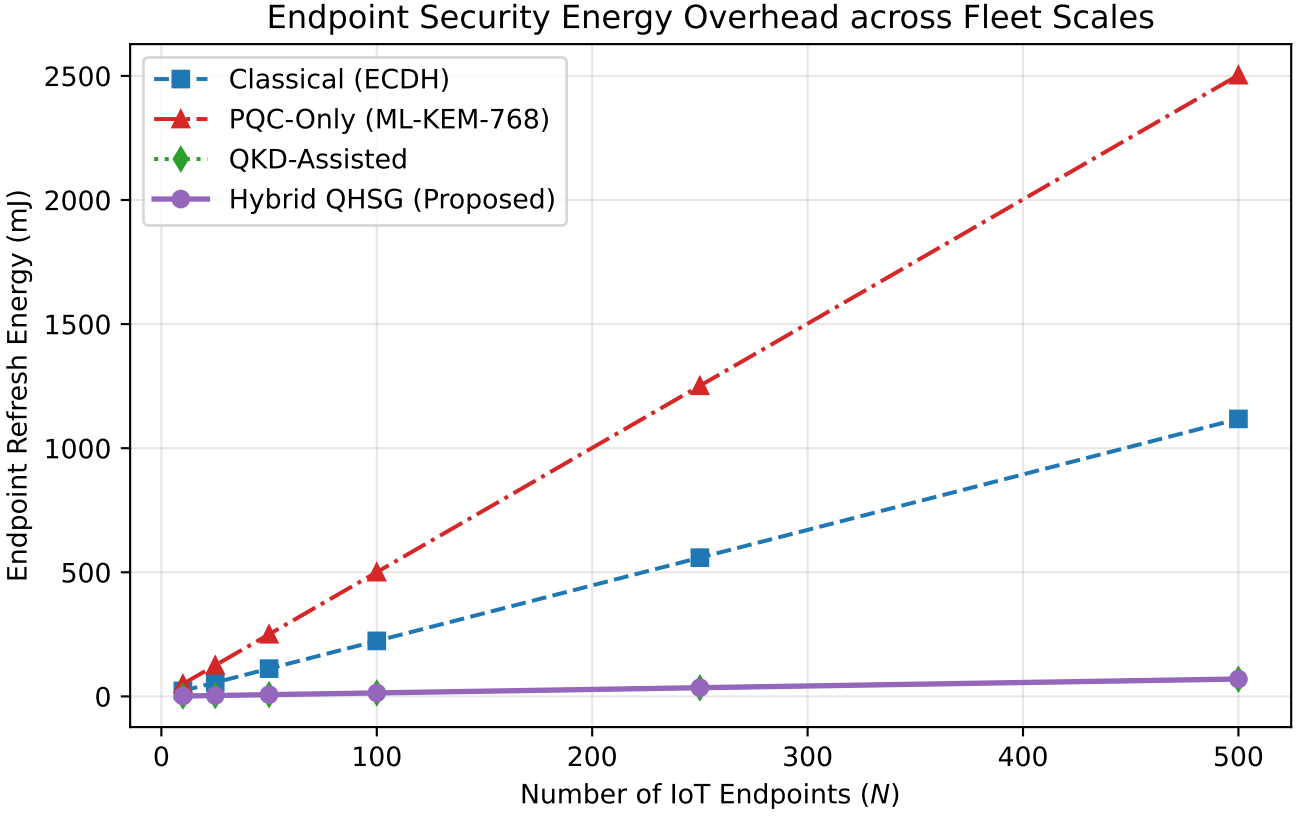
Endpoint Security Energy Overhead across Fleet Scales
Classical (ECDH)
PQC-Only (ML-KEM-768)
QKD-Assisted
Hybrid QHSG (Proposed)
Endpoint Refresh Energy (mJ)
2500
2000
1500
1000
500
0
0
100
200
300
400
500
Number of IoT Endpoints (N)